

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Bachelor of Technology (Civil Engineering) SEMESTER - 6 Summer 2025 ( Regular )**

**Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology**

**Semester : SEMESTER - 6**

**Subject Code & Name: BTCVOE605K - DEVELOPMENT ENGINEERING**

**Time : 3 hours]**

**[Total Marks : 60**

**Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

**Q1. Objective type questions. (Compulsory Question)**

**12**

- 1 Which of the following is NOT a core focus area in development engineering?
  - A. Socioeconomic development
  - B. Urban and rural development
  - C. Aerospace engineering
  - D. Scientific social research
- 2 Which of the following is a green building certification system in India?
  - A. BREEAM
  - B. GRIHA
  - C. ISO 9001
  - D. ASTM
- 3 What is the purpose of zoning in town planning?
  - A. To increase housing prices
  - B. To control land use and building density
  - C. To promote private businesses only
  - D. To eliminate public transport
- 4 What is the primary use of GIS in planning?
  - A. Entertainment
  - B. Web design
  - C. Mapping and layout preparation
  - D. Cooking
- 5 Which of the following best defines a 'research problem' in development engineering?
  - A. Any random issue in a community
  - B. A well-defined issue identified through fieldwork needing a development-based solution
  - C. An academic theory without practical implications
  - D. A financial audit of urban corporations

- 6 LEED certification is associated with which of the following?  
 A. Legal infrastructure  
 B. Road quality  
 C. Green building design and sustainability  
 D. Urban crime control
- 7 Which organization is responsible for energy standards often applied in sustainable building design?  
 A. ASHRAE  
 B. WHO  
 C. NASA  
 D. IMF
- 8 Which of the following is NOT typically part of a smart city infrastructure?  
 A. e-Governance  
 B. Urban mobility  
 C. Space missions  
 D. Waste management
- 9 Which element is considered a key performance benchmark in smart cities globally?  
 A. Population density  
 B. Amount of vehicles  
 C. ICT-based citizen services  
 D. Size of shopping malls
- 10 Which of these is a major challenge in rural public transport development?  
 A. Too many private airlines  
 B. Lack of demand  
 C. Sparse population and poor connectivity  
 D. High-speed train congestion
- 11 People's participation in rural development is essential because:  
 A. It reduces government funding  
 B. It ensures that local needs are addressed effectively  
 C. It delays project completion  
 D. It limits innovation
- 12 Which of the following is an interdisciplinary application of Geoinformatics?  
 A. Weather prediction  
 B. Land-use mapping  
 C. Disaster management  
 D. All of the above
- Q2. Solve the following.  
 A) Explain necessity and issues related to Development Engineering. 6  
 B) What is socio- economic development? Explain the factors effecting development. 6
- Q3. Solve the following.  
 A) Explain any one standard for sustainable communities and certification for green building? 6  
 B) Explain the formulation of research problems and explain various methods of data collection? 6
- Q4. Solve Any Two of the following.

- A) Explain the objectives of smart city planning and also explain various Smart Solution? 6
- B) What is geoinformatics and explain the benefits and limitations of Geoinformatics? 6
- C) Explain master plan and density control? 6
- Q5. Solve Any Two of the following.
- A) Explain District Planning and its functions. 6
- B) What are the infrastructure elements of smart city planning? Explain dimensions of smart cities. 6
- C) Illustrate the Problems faced by the villagers pertaining to various sectors or existing schemes. 6
- Q6. Solve Any Two of the following.
- A) Explain the concept of sustainable community and its development? 6
- B) Explain different levels of town planning? 6
- C) Explain the application of geoinformatics? 6

\*\*\* End \*\*\*